

Customer Segmentation & Behavior Analysis

Business/Data Analytics Portfolio Project

Tools & Technologies Used

Tool	Purpose
Excel	Data cleaning, exploratory analysis, pivot analysis
SQL (MySQL)	Customer metrics, KPI analysis, RFM preparation
Python	RFM analysis and customer segmentation
Power BI	Interactive dashboard and business storytelling

Project Objective

The objective of this project is to analyze customer purchasing behavior using transactional retail data and identify meaningful customer segments through RFM (Recency, Frequency, Monetary) analysis. The project focuses on deriving actionable business insights related to customer value, sales performance, purchasing trends, and customer retention opportunities.

Dataset Overview

The dataset contains retail transaction records including invoice details, product information, customer IDs, quantity purchased, unit prices, transaction dates, and country information.

The dataset was cleaned and transformed before analysis to ensure accuracy, consistency, and reliable analytical results.

Project Workflow

- **Excel Analysis:** Performed data cleaning using Power Query, created pivot tables, trend analysis, exploratory charts, slicers, and preliminary business insights.
- **SQL Analysis:** Validated transactional data, generated customer-level KPIs, performed RFM feature engineering, and derived customer behavior insights.
- **Python Analytics:** Conducted advanced RFM analysis and categorized customers into Premium, Regular, and Occasional customer groups.
- **Power BI Dashboard:** Developed an interactive dashboard with KPI cards, trend analysis, country-wise revenue insights, customer analysis, and slicer-based filtering.

Data Cleaning & Preparation

- Removed duplicate records
- Removed missing customer IDs
- Excluded cancelled transactions
- Filtered invalid quantity and unit price values
- Converted invoice date into datetime format
- Created Total Revenue column

Key KPIs Analyzed

KPI	Description
Total Revenue	Overall revenue generated
Total Orders	Total distinct customer orders
Total Customers	Unique customer count
Average Order Value	Average revenue per order
Purchase Frequency	Number of orders placed by customers
Monetary Value	Total customer spending
Recency	Days since last purchase

RFM Analysis

RFM analysis is a customer segmentation technique based on Recency, Frequency, and Monetary value.

Metric	Meaning
Recency	How recently a customer made a purchase
Frequency	How often a customer purchases
Monetary	How much money a customer spends

Customer Segmentation Logic

Segment	Description
Premium Customers	High spending and high purchase frequency
Regular Customers	Moderate spending behavior
Occasional Customers	Low purchase frequency and lower spending

Power BI Dashboard Features

- Total Revenue KPI
- Total Orders KPI
- Total Customers KPI
- Average Order Value KPI
- Monthly Revenue Trend

- Unit Price Trend
- Country-wise Revenue Analysis
- Top Customer Analysis
- Interactive slicers and filters

Business Insights

- Revenue demonstrates strong seasonal growth during peak purchasing months.
- A small percentage of customers contributes a major portion of total revenue.
- Premium customers exhibit significantly higher purchase frequency and spending behavior.
- Sales growth is primarily driven by higher order volume rather than pricing changes.
- Revenue contribution is concentrated within a limited number of countries.
- Repeat customers generate stronger long-term business value than one-time buyers.
- High-demand periods may require improved inventory planning and operational forecasting.
- Product purchase patterns indicate concentration around a limited set of high-performing products.

Final Conclusion

This project successfully combined Excel, SQL, Python, and Power BI to perform end-to-end customer behavior analysis and segmentation.

The analysis identified valuable customer groups, purchasing trends, and revenue-driving patterns that can support customer retention strategies, personalized marketing, revenue optimization, and business decision-making.

The project demonstrates practical analytical skills in data cleaning, SQL querying, RFM analysis, customer segmentation, dashboard development, and business insight generation.

Skills Demonstrated

- Data Cleaning & Transformation
- SQL Query Writing
- RFM Analysis
- Customer Segmentation
- Business Intelligence
- Power BI Dashboarding
- Analytical Thinking
- Business Insight Generation
- Data Visualization

Project Deliverables

File	Description
Excel Analysis File	Data cleaning and exploratory analysis
SQL Script File	Customer metrics and analytical queries
Python	RFM analysis and segmentation
Power BI Dashboard	Interactive business dashboard